

Day 3 Plan

Maths: Profit, Loss & Discount · Reasoning: Analogy · Science: Physics 3: Work, Energy, Power & Gravitation

Today's schedule

#	Block	Time	What you do	Where
0	Day 2 recall	15 min	10 questions from Day 2, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	Read the formulas & methods (10 min) → 2-3 book questions per pattern, timed → log mistakes	Section A
-	Break	10 min	Away from the screen	-
2	Reasoning	50 min	Learn the method (10 min) → 15 timed questions from your book → log	Section B
-	Break	10 min	Away from the screen	-
3	Science	55 min	Lessons with quick checks → 20-question practice set → answers	Section C
4	Current affairs	20 min	20 MCQs from the monthly PDF	Your PDF
5	Recall	10 min	Redo today's wrong questions; write 3 rules for tomorrow	Mistake log

Speed targets: Maths 60-70 sec per question · Reasoning 30-40 sec · Science 15-20 sec.

Block 0 — Day 2 recall (answers on the last page)

#	Question	Your answer
1	Newton's first law is also called the law of ...?	
2	Mass is the measure of ...?	
3	SI unit of momentum?	
4	Force needed to give a 5 kg mass an acceleration of 2 m/s ² ?	
5	Why does a cricketer pull the hands back while catching?	
6	A rocket works on conservation of ...?	
7	Which friction is the smallest?	
8	Formula for centripetal force?	
9	A person in a lift accelerating upward feels heavier or lighter?	
10	Maths: mean proportional of 4 and 16?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Profit, Loss & Discount

Core idea

Three prices: **CP** (cost price — what the seller paid), **MP** (marked/list price on the tag), and **SP** (selling price — what the buyer pays). Two rules solve most questions: **profit and loss % are always on CP**, and **discount % is always on MP**. Use 100 as the CP or MP whenever no value is given.

Formulas

Concept	Formula / rule
Profit / loss	Profit = SP – CP · Loss = CP – SP
Profit %	Profit/CP × 100 (always on CP)
SP from CP	SP = CP × (100 + P%)/100 · SP = CP × (100 – L%)/100
CP from SP	CP = SP × 100/(100 + P%) · CP = SP × 100/(100 – L%)
Discount	Discount = MP – SP · SP = MP × (100 – d)/100
Successive discounts a%, b%	Single equivalent = a + b – ab/100
Mark up x%, discount d%	Profit% = x – d – xd/100
Dishonest dealer (false weight)	Gain% = error/(true – error) × 100
Same SP, x% gain on one, x% loss on other	Always a net loss of x²/100 %
Buy x, get y free	Effective discount = y/(x + y) × 100
CP of x items = SP of y items	Profit% = (x – y)/y × 100

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Basic profit %	Profit ÷ CP × 100	CP ₹400, SP ₹500 → profit 100 → 25%
2. Find CP from SP	CP = SP × 100/(100 + P%)	SP ₹690 at 15% profit → 690 × 100/115 = ₹600
3. Marked price & discount	SP = MP × (100 – d)/100	MP ₹800, 15% off → ₹680
4. Mark-up + discount	x – d – xd/100	Marked 40% above CP, 20% off → 40 – 20 – 8 = 12% profit (CP 100 → MP 140 → SP 112 ✓)
5. Successive discounts	a + b – ab/100	20% then 10% → 30 – 2 = 28%
6. False weights	error/(true – error) × 100	Uses 900 g for 1 kg → 100/900 × 100 = 11⅓%
7. Same SP, gain & loss	Loss = x ² /100 %	Two items, one at 10% gain, one at 10% loss → 1% loss
8. Buy x get y free	y/(x + y) × 100	Buy 4 get 1 free → 1/5 → 20% discount
9. Profit as % of SP	Take SP = 100, find CP	Profit = 20% of SP → SP 100, CP 80 → 20/80 = 25% on CP
10. CP of x = SP of y	(x – y)/y × 100	CP of 12 = SP of 10 → 2/10 × 100 = 20% profit

Traps

- Profit % is on **CP**, not SP; discount % is on **MP**, not CP.
- Successive discounts of 20% + 10% are **not** 30% — they're 28%.
- 'Same SP with x% gain and x% loss' is **always a loss** — never 'no profit, no loss'.

B · REASONING — Analogy

An analogy question gives a pair with a relationship (A : B) and asks you to find the pair with the **same relationship**. **Method:** make a short sentence for the first pair ("a **carpenter** uses a **saw**"), then test each option with the same sentence. If two options fit, make the sentence more specific.

Word relationships that repeat in RRB papers

Relationship	Examples
Worker : tool	Carpenter : saw · tailor : needle · surgeon : scalpel · farmer : plough · painter : brush
Worker : product	Author : book · poet : poem · cobbler : shoes · sculptor : statue
Product : raw material	Paper : pulp · cloth : yarn/fibre · furniture : wood · butter : milk · glass : sand
Quantity : unit	Length : metre · current : ampere · force : newton · power : watt · resistance : ohm
Instrument : measures	Barometer : pressure · hygrometer : humidity · odometer : distance · seismograph : earthquakes
Animal : young one	Cow : calf · dog : puppy · cat : kitten · horse : foal · sheep : lamb · lion : cub · goat : kid
Animal : home	Bee : hive/apiary · horse : stable · dog : kennel · lion : den · bird : nest · pig : sty · spider : web
Male : female	Horse : mare · lion : lioness · dog : bitch · bull : cow · drake : duck · gander : goose
Individual : group	Fish : shoal · cattle : herd · bees : swarm · sheep : flock · wolves : pack · lions : pride · ships : fleet
Others	Country : capital/currency · part : whole (page : book) · synonym · antonym · cause : effect

Number and letter analogies

Test these rules in order: n^2 , n^3 , $n^2 \pm 1$, $n^3 \pm n$, $n \times (n + 1)$, sum or product of digits. For letters, write the positions (A = 1 ... Z = 26) and look at the gaps, exactly as in yesterday's alphabet series.

Patterns, with solved examples

Pattern	Example	Solution
Worker : tool	Doctor : stethoscope :: carpenter : ?	Tool used → saw
Young one	Cow : calf :: horse : ?	Foal
Unit	Current : ampere :: resistance : ?	Ohm
Group	Fish : shoal :: wolves : ?	Pack
Cube	3 : 27 :: 4 : ?	$n^3 \rightarrow$ 64
$n^2 - 1$	5 : 24 :: 7 : ?	$25 - 1$, so $49 - 1 =$ 48
$n(n + 1)$	6 : 42 :: 8 : ?	$6 \times 7 = 42$, so $8 \times 9 =$ 72
Letter shift	ACE : BDF :: GIK : ?	Each letter +1 → HJL
Opposite letters	AZ : BY :: CX : ?	Next opposite pair → DW

Speed rule

Say the relationship as a full sentence. 'Fish live in a shoal' is vaguer than 'a group of fish is called a shoal' — the second one picks the right answer.

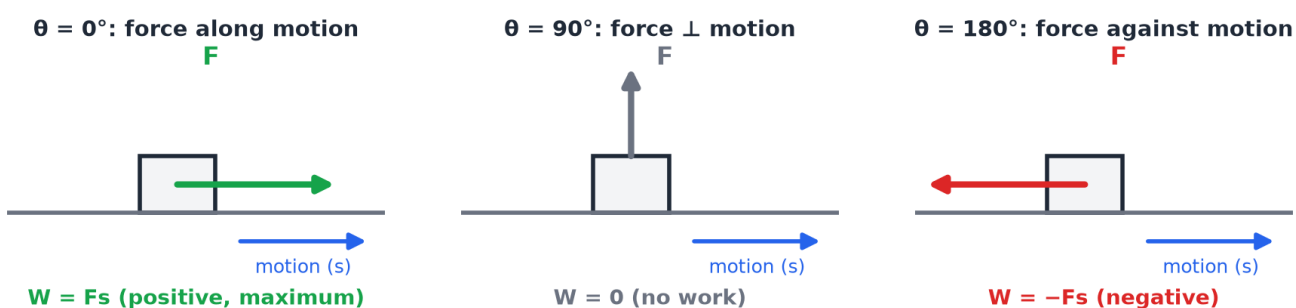
C · SCIENCE — Physics 3: Work, Energy, Power & Gravitation

9 lessons. Day 2 was about forces. Today: what forces achieve (work), what that costs (energy), how fast it's done (power), and the force that holds the universe together (gravitation).

Lesson 1 · Work — the physics meaning (about 6 min)

In everyday language, 'work' means any effort. In physics, **work is done only when a force moves something**. Push a wall all day and you'll be tired, but you've done **zero work** — the wall didn't move.

$W = F \times s \times \cos \theta$, where θ is the angle between the force and the direction of motion. Unit: **joule (J)**. Work is a **scalar**.



Case	Example	Work
Force along the motion	Pulling a cart forward	Positive
Force at 90° to the motion	Carrying a bag while walking on level ground · centripetal force in circular motion · a satellite's gravity in a circular orbit	Zero
Force against the motion	Friction on a sliding box · gravity on a ball thrown upward · brakes	Negative
No displacement	Pushing a wall · holding a heavy bag still	Zero

Worked example — work

10 N pulls a box 5 m along the force: $W = 10 \times 5 \times \cos 0^\circ = \mathbf{50\ J}$.

Same force at 60° to the motion: $W = 10 \times 5 \times \cos 60^\circ = 10 \times 5 \times 0.5 = \mathbf{25\ J}$.

Exam trap

A coolie carrying a load on his head and walking on a level platform does **zero work against gravity** (the force is vertical, the motion is horizontal).

Quick check 1 — answer out loud before moving on (answers at the end)

a) Is any work done in holding a 20 kg bag still for 10 minutes? b) What is the work done by the centripetal force in circular motion?

Lesson 2 · Energy and kinetic energy (about 6 min)

Energy is the capacity to do work. Same unit as work: joule. Energy comes in many forms — kinetic, potential, heat, light, sound, chemical, electrical, nuclear.

Kinetic energy (KE) is the energy an object has **because it's moving**: $KE = \frac{1}{2} m v^2$.

Because v is squared, speed matters much more than mass: **double the speed → 4 times the KE**; triple → 9 times.

Worked example — kinetic energy

A 2 kg ball at 3 m/s: $KE = \frac{1}{2} \times 2 \times 9 = \mathbf{9\ J}$.

At 6 m/s (double): $KE = \frac{1}{2} \times 2 \times 36 = \mathbf{36\ J} = 4 \times 9\ \checkmark$

Link with momentum: $KE = p^2/2m$. So if the momentum doubles (same mass), KE becomes 4 times. If two bodies have the **same KE**, the **heavier** one has **more momentum** ($p = \sqrt{2m \times KE}$).

Quick check 2 — answer out loud before moving on (answers at the end)

a) If a car's speed is tripled, its KE becomes how many times? b) What is the KE of a 4 kg body moving at 5 m/s?

Lesson 3 · Potential energy (about 3 min)

Potential energy (PE) is **stored** energy, due to position or shape. Lift a book onto a shelf: the work you did is stored as gravitational PE, ready to be released if it falls.

Gravitational PE = $m g h$ (h = height above a chosen reference level). Other stored energy: a stretched bow or rubber band, a compressed spring, a wound clock spring (elastic PE), water stored in a dam.

Worked example — PE

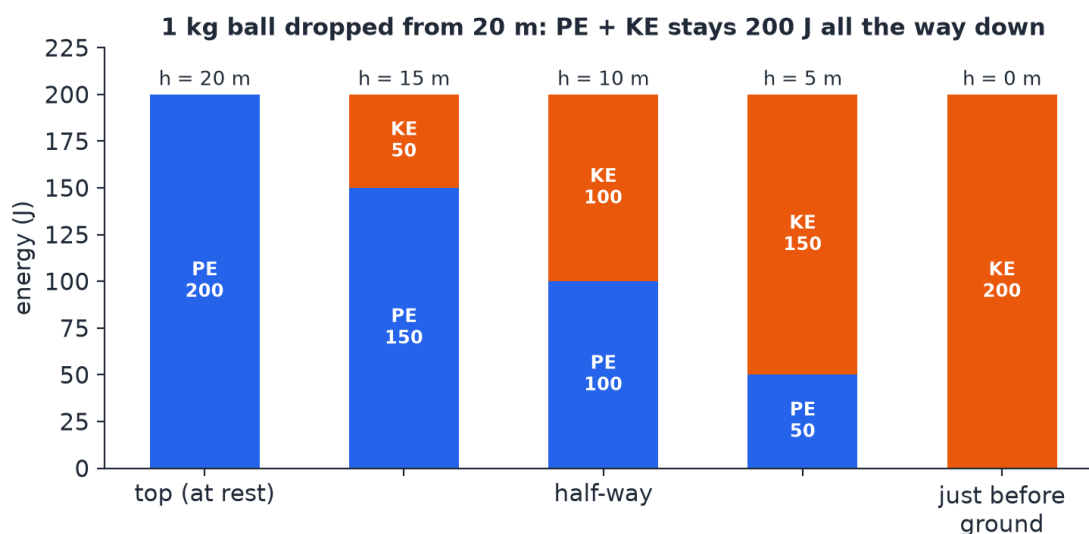
A 5 kg box lifted 10 m ($g = 10$): $PE = 5 \times 10 \times 10 = \mathbf{500\ J}$.

Quick check 3 — answer out loud before moving on (answers at the end)

What is the PE of a 2 kg book on a 3 m high shelf ($g = 10$)?

Lesson 4 · Conservation of energy and energy conversions (about 6 min)

Law of conservation of energy: energy can be neither created nor destroyed; it only **changes from one form to another**. The total stays the same.



As the ball falls, PE turns into KE, but PE + KE stays 200 J. A swinging pendulum does the same: maximum PE at the ends, maximum KE at the lowest point.

Worked example — speed after falling

Dropped from 20 m: all PE becomes KE $\rightarrow mgh = \frac{1}{2}mv^2 \rightarrow v = \sqrt{2gh} = \sqrt{400} = \mathbf{20 \text{ m/s}}$ (the same as Day 1's $v^2 = 2gh$).

Device / process	Energy conversion
Electric motor, fan	Electrical $\rightarrow$ mechanical
Generator / dynamo	Mechanical $\rightarrow$ electrical
Battery / cell	Chemical $\rightarrow$ electrical
Solar cell	Light $\rightarrow$ electrical
Microphone	Sound $\rightarrow$ electrical
Loudspeaker	Electrical $\rightarrow$ sound
Electric bulb	Electrical $\rightarrow$ light (+ heat)
Electric iron / heater	Electrical $\rightarrow$ heat
Steam engine	Heat $\rightarrow$ mechanical
Photosynthesis	Light $\rightarrow$ chemical

Quick check 4 — answer out loud before moving on (answers at the end)

a) What energy conversion happens in a dynamo? b) And in a microphone?

Lesson 5 • Power (about 4 min)

Power = how fast work is done (or energy is used): $P = W / t$. Unit: **watt (W)** = 1 J/s. Two people climbing the same stairs do the same work, but the faster one has more power. **1 HP = 746 W**. The electricity 'unit' is 1 kWh = 3.6×10^6 J (Day 1).

Worked example — climbing stairs

A 50 kg person climbs 10 m of stairs in 10 s ($g = 10$).

Work = $mgh = 50 \times 10 \times 10 = 5000$ J. Power = $5000 \div 10 = \mathbf{500\ W}$.

Another form: $P = F \times v$. A car engine pushing with 2000 N at 20 m/s delivers $2000 \times 20 = 40,000$ W = **40 kW**.

Quick check 5 — answer out loud before moving on (answers at the end)

A machine does 1500 J of work in 5 s. What is its power?

Lesson 6 • Newton's law of gravitation (about 5 min)

Every object in the universe attracts every other object. The force is larger for bigger masses and falls quickly with distance:

$F = G m_1 m_2 / r^2$ · $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ — the **universal gravitational constant** (same everywhere in the universe).



$F = G m_1 m_2 / r^2$ — **double the distance, the force becomes $\frac{1}{4}$**

This is an **inverse-square law**: double the distance $\rightarrow \frac{1}{4}$ of the force; triple $\rightarrow \frac{1}{9}$. Double one mass $\rightarrow$ double the force. Gravity explains falling objects, the Moon's orbit, planets' orbits and **ocean tides** (mainly caused by the Moon's pull).

Exam trap

G is a universal constant (6.67×10^{-11}). **g** is the acceleration due to gravity ($\approx 9.8 \text{ m/s}^2$) and it **changes** from place to place. Don't mix them up.

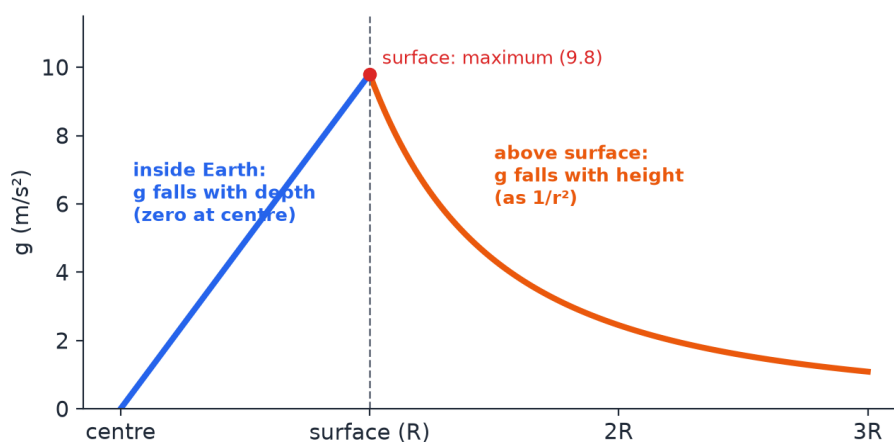
Quick check 6 — answer out loud before moving on (answers at the end)

The distance between two masses is halved. What happens to the gravitational force between them?

Lesson 7 · Acceleration due to gravity (g) and how it changes

(about 5 min)

g is the acceleration of a freely falling body near Earth: about **9.8 m/s²**, given by $g = GM/R^2$ (M, R = Earth's mass and radius). It does **not** depend on the mass of the falling object.



Where	g is...	Why
Poles vs equator	Maximum at the poles, minimum at the equator	Earth bulges at the equator (larger radius) and its rotation reduces the effective g there
Going up (height)	Decreases	Further from Earth's centre
Going down (depth)	Decreases	Less of Earth's mass is 'below' you
Centre of the Earth	Zero	Pulled equally in all directions
On the Moon	About 1/6 of Earth's	Much smaller mass

Quick check 7 — answer out loud before moving on (answers at the end)

a) Where on Earth is g largest? b) What is g at the centre of the Earth?

Lesson 8 · Mass vs weight (about 4 min)

	Mass	Weight
Meaning	Amount of matter in a body	Force with which Earth pulls the body: W = mg
Unit	kg	newton (N)
Type	Scalar	Vector (acts downward)
Changes from place to place?	Never	Yes — changes with g
Measured with	Beam (two-pan) balance	Spring balance
In free fall / orbit	Same	Apparent weight = 0 (weightless)

Worked example — on the Moon

A 60 kg person: mass on the Moon = **60 kg** (unchanged).

Weight on Earth = $60 \times 9.8 = 588$ N; on the Moon = $588 \div 6 = \mathbf{98\text{ N}}$.

Quick check 8 — answer out loud before moving on (answers at the end)

A body has a mass of 12 kg on Earth. What are its mass and its weight on the Moon (g on Earth = 10)?

Lesson 9 · Escape velocity, satellites and Kepler's laws (about 7 min)

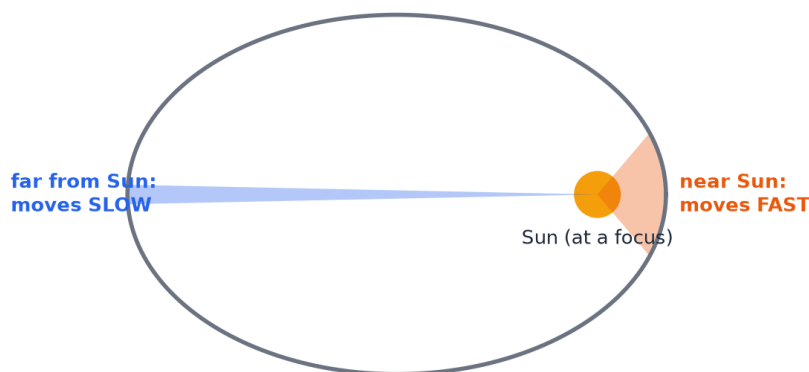
Escape velocity = the minimum speed needed to escape a planet's gravity completely. Earth: **11.2 km/s**. Moon: about **2.4 km/s**. It does **not** depend on the mass of the object being launched.

Why the Moon has no atmosphere: its escape velocity is so low that gas molecules, moving fast because of heat, escape into space.

Satellite type	Height / period	Use
Low Earth orbit (near the surface)	Orbital speed ≈ 7.9 km/s · period ≈ 84 min close to the surface	Remote sensing, the ISS
Geostationary	$\approx 36,000$ km up · period 24 hours · moves west $\rightarrow$ east above the equator, so it looks fixed in the sky	Communication, TV broadcast
Polar	Lower orbit passing over the poles	Weather, mapping, remote sensing

Kepler's three laws of planetary motion

Equal areas are swept in equal times



Law	Statement	Plain meaning
1. Orbits	Planets move in ellipses with the Sun at one focus	Orbits are slightly oval, not perfect circles
2. Areas	The line Sun-planet sweeps equal areas in equal times	A planet moves fastest when nearest to the Sun
3. Periods	$T^2 \propto r^3$	Planets further away take much longer to go round

Quick check 9 — answer out loud before moving on (answers at the end)

a) What is the escape velocity from Earth? b) What is the period of a geostationary satellite? c) When does a planet move fastest?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Work	$W = Fs \cos\theta$ · joule · scalar · zero when $\theta = 90^\circ$ (coolie on level ground, circular motion) or no displacement · negative for friction
Kinetic energy	$KE = \frac{1}{2}mv^2$ · double the speed $\rightarrow 4\times KE$ · $KE = p^2/2m$
Potential energy	$PE = mgh$ · stored energy (bow, spring, dam)
Conservation	Energy only changes form · falling body: $PE + KE$ constant · $v = \sqrt{2gh}$
Conversions	Motor: elec $\rightarrow$ mech · dynamo: mech $\rightarrow$ elec · cell: chem $\rightarrow$ elec · solar cell: light $\rightarrow$ elec · mic: sound $\rightarrow$ elec · speaker: elec $\rightarrow$ sound
Power	$P = W/t = Fv$ · watt · 1 HP = 746 W · 1 kWh = 3.6 MJ
Gravitation	$F = Gm_1m_2/r^2$ · $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ (universal) · inverse square · tides
g	$\approx 9.8 \text{ m/s}^2$ · max at the poles, min at the equator · decreases with height and depth · zero at the centre · Moon $\approx g/6$
Mass vs weight	Mass constant (kg, balance) · weight = mg (N, spring balance) · Moon weight = $1/6$
Escape velocity	Earth 11.2 km/s · Moon 2.4 km/s (so no atmosphere) · independent of the object's mass
Satellites	Geostationary: 24 h, $\approx 36,000$ km, equatorial, communication · polar: weather/mapping
Kepler	Elliptical orbits · equal areas in equal times (fastest near the Sun) · $T^2 \propto r^3$

Science practice set — 20 questions · target 6 minutes

#	Question		
1	Work done when the force is perpendicular to the displacement is: (a) maximum (b) negative (c) infinite (d) zero		
2	The SI unit of work is: (a) watt (b) newton (c) joule (d) pascal		
3	A coolie carrying a load on his head walks on a level road. Work done against gravity is: (a) positive (b) zero (c) negative (d) maximum		
4	If the speed of a body is doubled, its kinetic energy becomes: (a) 2 times (b) half (c) unchanged (d) 4 times		
5	The kinetic energy of a 2 kg body moving at 3 m/s is: (a) 6 J (b) 18 J (c) 3 J (d) 9 J		
6	The potential energy of a 5 kg body at a height of 10 m ($g = 10 \text{ m/s}^2$) is: (a) 50 J (b) 5000 J (c) 150 J (d) 500 J		
7	An electric motor converts: (a) mechanical energy into electrical energy (b) chemical energy into electrical energy (c) heat energy into mechanical energy (d) electrical energy into mechanical energy		
8	A dynamo converts: (a) mechanical energy into electrical energy (b) electrical energy into mechanical energy (c) chemical energy into mechanical energy (d) light energy into electrical energy		
9	1 horsepower is equal to: (a) 1000 W (b) 746 kW (c) 746 W (d) 550 W		
10	A 50 kg person climbs 10 m in 10 s ($g = 10 \text{ m/s}^2$). The power developed is: (a) 500 W (b) 50 W (c) 5000 W (d) 250 W		
11	If the distance between two masses is doubled, the gravitational force becomes: (a) half (b) double (c) four times (d) one-fourth		
12	The value of the universal gravitational constant G is: (a) 9.8 m/s^2 (b) $6.67 \times 10^{11} \text{ N m}^2 \text{ kg}^{-2}$ (c) $3 \times 10^8 \text{ m/s}$ (d) $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$		
13	The acceleration due to gravity is maximum at the: (a) poles (b) equator (c) centre of the Earth (d) top of a mountain		
14	The value of g at the centre of the Earth is: (a) 9.8 m/s^2 (b) infinite (c) zero (d) 4.9 m/s^2		
15	The weight of a body on the Moon compared to that on Earth is: (a) the same (b) six times (c) one-sixth (d) zero		
16	The mass of a body on the Moon compared to that on Earth is: (a) the same (b) one-sixth (c) six times (d) zero		
17	The escape velocity from the Earth's surface is about: (a) 7.9 km/s (b) 11.2 km/s (c) 2.4 km/s (d) $3 \times 10^8 \text{ m/s}$		
18	The period of a geostationary satellite is: (a) 24 hours (b) 12 hours (c) 365 days (d) 1 hour		
19	The Moon has no atmosphere because: (a) it is far from the Sun (b) it has no water (c) its escape velocity is low (d) it does not rotate		
20	According to Kepler's second law, a planet moves fastest when it is: (a) farthest from the Sun (b) closest to the Sun (c) at the mid-point of its orbit (d) at the same speed everywhere		
Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	d	Lesson 1
2	c	Lesson 1
3	b	Force vertical, motion horizontal (Lesson 1)
4	d	$KE \propto v^2$ (Lesson 2)
5	d	$\frac{1}{2} \times 2 \times 9$ (Lesson 2)
6	d	mgh (Lesson 3)
7	d	Lesson 4
8	a	Lesson 4
9	c	Lesson 5
10	a	$5000 \text{ J} \div 10 \text{ s}$ (Lesson 5)
11	d	Inverse square (Lesson 6)
12	d	Lesson 6
13	a	Lesson 7
14	c	Lesson 7
15	c	Lesson 8
16	a	Mass never changes (Lesson 8)
17	b	Lesson 9
18	a	Lesson 9
19	c	Lesson 9
20	b	Lesson 9

Quick checks

QC	Answers
1	a) No — no displacement, so $W = 0$ b) zero (force $\perp$ motion)
2	a) 9 times b) $\frac{1}{2} \times 4 \times 25 = 50 \text{ J}$
3	$2 \times 10 \times 3 = 60 \text{ J}$
4	a) mechanical $\rightarrow$ electrical b) sound $\rightarrow$ electrical
5	$1500 \div 5 = 300 \text{ W}$
6	It becomes 4 times (r halved $\rightarrow 1/(\frac{1}{2})^2 = 4$)
7	a) at the poles b) zero
8	Mass = 12 kg (unchanged); weight on Earth = 120 N $\rightarrow$ on the Moon = 20 N
9	a) 11.2 km/s b) 24 hours c) when it's closest to the Sun

Day 2 recall (Block 0)

#	Answer	#	Answer
1	Inertia	6	Momentum
2	Inertia	7	Rolling
3	kg·m/s	8	mv^2/r
4	10 N	9	Heavier: $R = m(g + a)$
5	More time → less force (impulse)	10	$\sqrt{64} = 8$

Send me tonight

Day 2 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.